

THE BOND MARKET — COMPLETE REFERENCE

Mechanics · Yield · Risk · Sectors · Portfolio Strategy · Connection to Stocks & Everyday Life

① BOND BASICS — WHAT YOU ARE ACTUALLY DOING

The Transaction	When you buy a bond, you are lending money to the issuer (government, corporation, or municipality). In return, the issuer promises to pay you periodic interest (the coupon) and return your principal at maturity .
Coupon	The fixed annual interest rate paid on the bond's face value. A \$1,000 bond with a 4% coupon pays \$40/year (\$20 semi-annually). The coupon rate never changes — but your effective yield does based on what you paid.
Par / Face Value	The original loan amount — typically \$1,000 . What the issuer pays you back at maturity regardless of what you paid for the bond in the market.
Maturity	When the issuer returns your principal. T-Bills : under 1yr T-Notes : 2–10yr T-Bonds : 30yr. Longer maturity = more interest rate risk = higher yield demanded.
Golden Rule	Price and Yield ALWAYS move in opposite directions . Buy a bond below par → yield rises above coupon. Buy above par → yield falls below coupon. This is the single most important mechanic in bond investing.
Who Issues?	• U.S. Treasury — safest, benchmark for all other bonds • Corporations — investment grade (BBB- and above) or high yield / "junk" (below BBB-) • Municipalities — state/local gov't, often tax-exempt income • Foreign Governments — sovereign debt, adds currency risk

② MEASURING YIELD — WHAT YOUR RETURN ACTUALLY IS

Current Yield	Coupon ÷ Price . Simplest measure. If a \$1,000 bond pays \$40/yr but you bought it for \$950, current yield = 4.21%. Used for preferred stocks and perpetual securities.
Yield to Maturity (YTM)	The standard yield measure for bonds . Accounts for the coupon payments, time value of money, AND the gain/loss from buying at a discount/premium versus par. It's the internal rate of return on the bond. A discount bond's YTM exceeds its coupon; a premium bond's YTM is below its coupon.
Yield to Call (YTC)	For callable bonds — same as YTM but uses the call date instead of maturity. If a bond is priced above par, the issuer may call it early (redeem at par) to refinance at lower rates, hurting the investor.
Yield to Worst (YTW)	The lower of YTC and YTM . The most conservative and realistic yield measure for callable bonds. Always use YTW when evaluating callable bonds — it's what you're truly guaranteed at minimum.
Basis Points	Bond market unit of measurement. 1 basis point (bp) = 0.01% . A move from 3.00% to 3.50% = 50 basis points. Fed decisions are typically described in 25bp or 50bp increments.
Real Yield	Nominal yield minus inflation . If the 10yr pays 4.5% but inflation is 3%, your real yield is 1.5%. Negative real yields (inflation exceeds nominal yield) historically boost gold and risk assets. Positive real yields tighten financial conditions.

③ THE YIELD CURVE — THE MARKET'S MOST POWERFUL SIGNAL

Shape	What It Looks Like	What It Means
Normal	Long-term yields > short-term	Healthy economy, investors expect growth, compensated for locking up money longer
Flat	Short and long yields nearly equal	Economy at inflection point — uncertainty, often precedes inversion
Inverted	Short-term yields > long-term	Recession warning — investors buying long bonds expecting future rate cuts. Has preceded every U.S. recession since the 1970s

Steepening	Long rates rising faster than short	Market raising inflation expectations for future years; or early recovery from recession
Flattening	Short rates rising faster than long	Market expects Fed to raise short-term rates soon; or long-run inflation expectations falling

Key Spread to Watch: ^TNX (10yr) minus ^IRX (3mo). When this goes **NEGATIVE** = inversion. The un-inversion is typically when the recession actually arrives — the curve inverts as a warning, then un-inverts as the economy breaks.

④ COMPENSATION FOR RISK — WHY DIFFERENT BONDS PAY DIFFERENT YIELDS

Credit Risk	The risk the issuer defaults. Higher default probability = higher yield demanded. U.S. Treasuries are the benchmark (lowest yield, most credit-safe). All other bonds are measured by their spread over Treasuries .
Interest Rate Risk	Longer maturities = more price sensitivity to yield changes. A 1% rise in yield cuts a 2yr bond price ~1%, but cuts a 30yr bond price ~17%. This is why long bonds pay more — you need compensation for that volatility.
Duration	Measures price sensitivity to yield changes. Duration of 6 = a 1% yield rise drops price ~6%. Higher duration bonds (longer maturity, lower coupon) are most sensitive. TLT has very high duration — it moves dramatically with rate changes.
Credit Spreads	The extra yield a bond pays over a Treasury of the same maturity. Investment grade corporates: ~100-200bps over Treasury. High yield: ~300-600bps+. Widening spreads = market fears defaults/recession. Tightening spreads = confidence. Watch LQD for this signal.
Credit Ratings	Investment Grade: AAA → BBB- (Moody's: Aaa → Baa3) — lower default risk High Yield / Junk: BB+ and below — higher default risk, higher yield Historical 10yr default rates: IG corporates ~2.8% HY corporates ~32.4%
Convexity	For standard bonds: the price gain from a yield decline exceeds the price loss from the same yield rise. This is positive convexity — it's a built-in advantage. Callable bonds and MBS can have negative convexity (bad for investors), which is why they pay higher yields.

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⑤ BOND MARKET SECTORS — WHAT EACH TYPE IS & WHEN IT PERFORMS

Sector	What It Is	Key Signal / Ticker	Performs Best When...
Treasuries	U.S. gov't debt, zero credit risk, AAA rated. Benchmark for all bonds.	^IRX, ^FVX, ^TNX, ^TYX, TLT	Recession fear, market stress — 'flight to quality'
TIPS	Treasury Inflation-Protected. Principal adjusts with CPI. Real yield instrument.	TIP ETF	Inflation rising — protects purchasing power of your return
MBS / GSEs	Mortgage-Backed Securities from Fannie/Freddie. You own a slice of home loan payments. Prepayment risk when rates fall.	MBB ETF	Stable rates — prepayment and extension risk hurt returns in volatile rate environments
Inv. Grade Corp	Debt from financially strong companies (BBB- and above). Spreads over Treasuries signal corporate health.	LQD ETF	Strong economy, tight spreads, low default risk expected
High Yield (Junk)	Below-investment-grade company debt. High default risk, high yield. Returns correlate more with stocks than Treasuries.	HYG / JNK ETFs	Economic expansion — investors willing to take credit risk for higher return
Municipals	State/local gov't bonds. Interest often exempt from federal taxes. Use taxable equivalent yield (TEY) for comparison.	MUB ETF	High-income investors seeking tax-free income; historically very low default rates
Emerging Mkt	Sovereign/corporate debt from developing countries. Dollar-denominated or local currency. Higher yields, higher risk.	EMB / EMLC ETFs	Risk-on environments, strong global growth, stable USD

⑥ BONDS vs. STOCKS — HOW THEY INTERACT

Asset	When Yields RISE	When Yields FALL
Growth Stocks	Sell off hardest — future earnings discounted more heavily	Rally hardest — cheap money + higher present value of future earnings
Value Stocks	Hold better — near-term earnings less affected by discount rate	Moderate rally — benefit but less than growth stocks
Bank Stocks	Often benefit — wider spread between lending and deposit rates	Compress — net interest margin narrows, less profit per dollar lent
REITs	Hurt badly — high debt loads + compete with bond yields for income	Rally — cheaper financing, income-seekers return to real estate
High Yield Bonds	Fall — spreads widen, default risk reprices. Behave more like stocks than Treasuries	Rally — tighter spreads, risk appetite returns
Long Treasuries (TLT)	Fall in price sharply — existing bonds less attractive vs new issues	Rise — flight to safety or rate cut pricing boosts price

Cash / T-Bills	More attractive — risk-free return competes with equities (TINA trade ends)	Less attractive — pushes capital back into risk assets
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7 HOW BONDS AFFECT YOUR EVERYDAY LIFE

Mortgage Rates	Tied directly to the 10yr (^TNX) and 30yr (^TYX) Treasury yield. A 1% rise in the 10yr can add hundreds of dollars/month to a new home payment. MBS pricing also factors in.
Car & Student Loans	Benchmarked to short and mid-term Treasury rates. Fed hikes ripple into your monthly payment on any new loan within weeks to months.
Savings / CD Rates	When the Fed raises rates, banks pay more on savings accounts and CDs. Your high-yield savings rate tracks the Fed Funds rate (^IRX) closely — this is the one time rising rates help regular savers.
Credit Card APR	Credit card rates float with the Prime Rate, which moves directly with Fed hikes. Every 25bp Fed increase typically adds to your credit card APR within one billing cycle.
Corporate Spending	Companies borrow via corporate bonds. Higher rates = costlier expansion, hiring, R&D.; This compresses future earnings and stock valuations — the core mechanism of Fed tightening.
Government Debt	The U.S. pays interest on ~\$34 trillion in debt. When yields rise, debt servicing costs increase — crowding out other spending and adding to the deficit. At 5% on \$34T, that's ~\$1.7T/year just in interest.
Municipal Projects	Higher rates make it more expensive for cities and states to issue bonds for roads, schools, and hospitals. Infrastructure investment slows. This is how Fed policy reaches every level of government.

8 BOND PORTFOLIO STRATEGIES — LADDERS, BARBELLS & TOTAL RETURN

Total Return	Coupon income + price change + reinvested coupon income. Not just yield — if you sell before maturity, price change matters. Rising yields hurt price but if held to maturity you always get par back (barring default).
Bond Ladder	Buy bonds with staggered maturities (e.g. 1yr, 5yr, 10yr). As each matures, reinvest at the longest rung. Benefit: smooths out rate risk — you're always reinvesting some portion at current rates. Protects against both rising and falling rates.
Barbell Strategy	Combine short maturities (2yr) + long maturities (10yr) , skip the middle. Short end: redeploy quickly if rates rise. Long end: captures yield curve steepness and benefits from 'rolling down the curve'. Outperforms when yield curve flattens.
Rolling Down the Curve	A 10yr bond becomes a 9yr bond after one year. If the yield curve is normally sloped, the bond's yield naturally falls (price rises) simply due to time passing — even with no rate change. This adds return on top of the coupon.
Funds vs. Individual Bonds	Individual bonds: predictable — hold to maturity, get par back. Price fluctuates but damage is temporary. Bond ETFs/Funds (e.g. TLT): maturity doesn't shorten over time, so rate risk is permanent. More flexible but less predictable for income-focused investors.
Duration Management	If you expect rates to rise → shorten duration (own shorter maturities, less price sensitivity). If you expect rates to fall → lengthen duration (own longer maturities like TLT — more price upside). Duration is your primary lever for managing interest rate risk.

BONDS MOVE FIRST	Bond traders are institutional & data-driven. Stocks are emotional. Watch bonds — stocks follow.	^TNX IS THE NORTH STAR	Every asset on Earth is priced relative to the 10-Year Treasury Yield. When it moves, everything reprices.	CONTEXT IS EVERYTHING	The same yield move can be bullish or bearish. The WHY behind the move determines the reaction.
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